Quick Sort

I/P :-

7	1	3	5	2	6	4
0	1	2	3	4	5	6

pivot = 4 (last element of the array)

1	3	2	4	7	5	6
0	1	2	3	4	5	6

left of 4

Right of 4

1	2	3	4	7	5	6
0	1	2	3	4	5	6

left of 2

right of 2

Left of 4

Right of 4

1	2	3	4	7	5	6
0	1	2	3	4	5	6

1	2	3	4	5	6	7
0	1	2	3	4	5	6

Left of 4 Sorted

left of 6

right of 6

Right of 4 Sorted

<u>O/P</u>:-

1	2	3	4	5	6	7
0	1	2	3	4	5	6

Pseudo Code for Quick Sort

```
QS(A, start, end)
  {
  if (A.length==1)
    {
    return ;
    }
pi = partition (A, start, end)
QS (A, start, pi-1)
QS (A, pi+1, end)
  }
```

Pseudocode for partition

```
Partition (A, start, end)
{
  pivot = A[end];
  pi = start;
for (i= start → i = end - 1; i++)
  {
   if (A[i] <= pivot)
    {
     swap A[pi] with A[i]
     pi++;
    }
  }
}
```

Swap A[pi] with A[end]

return pi ;

}

~~Time Complexity is~~ $\boxed{\cancel{O(n^2)}}$

Time Complexity

$$T(n) = 2T\left(\frac{n}{2}\right) + n \quad \text{— (1)}$$

$$T\left(\frac{n}{2}\right) = 2T\left(\frac{n}{4}\right) + \frac{n}{2} \quad \text{— (2)}$$

$$T\left(\frac{n}{4}\right) = 2T\left(\frac{n}{8}\right) + \frac{n}{4} \quad \text{— (3)}$$

Apply (2) in (1)

$$T(n) = 2\left[2T\left(\frac{n}{4}\right) + \frac{n}{2}\right] + n$$

$$= 4T\left(\frac{n}{4}\right) + \cancel{2} \cdot \frac{n}{\cancel{2}} + n$$

$$= 4T\left(\frac{n}{4}\right) + 2n \quad \text{—— } (4)$$

Apply eqn (3) in (4)

$$= 4\left[2T\left(\frac{n}{8}\right) + \frac{n}{4}\right] + 2n$$

$$= 8T\left(\frac{n}{8}\right) + \cancel{4} \cdot \frac{n}{\cancel{4}} + 2n$$

$$= 8T\left(\frac{n}{8}\right) + 3n$$

Generalize

$$= 2^i T\left(\frac{n}{2^i}\right) + in \quad \text{—— } (5)$$

Keep $\frac{n}{2^i} = 1$ — (8)

$\boxed{n = 2^i}$ — (6)

Apply log on both sides

$\log n = \log 2^i$

$\boxed{\log n = i}$ — $\cancel{(7)}$

Apply eqn (6) and (7), (8) in eqn (5)

$= n\,T(1) + \log n \cdot n$

$= n \cdot 1 + \log n \cdot n$

$= n + n \log n$

$\therefore$ Time complexity is $\boxed{O(n \log n)}$